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Arthur Cayley

848 (continued). On the Transformation of the Double-Theta Functions.

(Continuation of Article 45; printed page 381.)

and consider the function

$$\Pi(x, y) = \Theta(z_0 + Az_2 + Hz_3, z_1 + Hz_2 + Bz_3) \exp. i\pi\{(z_0z_2 + z_1z_3) + (A, H, B \mid z_2, z_3)^2\},$$

where Θ denotes the function

$$\Theta\left(\begin{smallmatrix} \mu, & \nu \\ q, & r \end{smallmatrix}\right)(A', H', B').$$

It is to be shown that Π is a function

$$\Pi\left(\begin{smallmatrix} \mu', & \nu' \\ q', & r' \end{smallmatrix}\right)(A', H', B'),$$

where the new parameters A', H', B' are given in terms of the original parameters A, H, B by the equations

$$1 : A : H : B : H^2 - AB = T : P : Q : R : S,$$

$$1 : A' : H' : B' : H'^2 - A'B' = T' : P' : Q' : R' : S',$$

and where μ', ν', q', r' have the values

$$\begin{aligned} \mu' &= \mu a_0 + \nu a_1 + qa_2 + ra_3 + a_0a_2 + a_1a_3, \\ \nu' &= \mu b_0 + \nu b_1 + qb_2 + rb_3 + b_0b_2 + b_1b_3, \\ q' &= \mu c_0 + \nu c_1 + qc_2 + rc_3 + c_0c_2 + c_1c_3, \\ r' &= \mu d_0 + \nu d_1 + qd_2 + rd_3 + d_0d_2 + d_1d_3; \end{aligned}$$

viz. it is to be shown that the function Π , as above defined, satisfies the fundamental equations

$$\begin{aligned} \Pi(x+1, y) &= (-)^\mu \Pi(x, y), \\ \Pi(x, y+1) &= (-)^\nu \Pi(x, y), \\ \Pi(x+A, y+H) &= (-)^q \Pi(x, y) \exp. -2\pi k(2x+A'), \\ \Pi(x+H, y+B) &= (-)^r \Pi(x, y) \exp. -2\pi k(2y+B'), \end{aligned}$$

and

$$\Pi(-x, -y) = (-)^{\mu'+\nu'+q'+r'} \Pi(x, y).$$

46. It is proper in the first place to show how it is that A', H', B' are capable of being the parameters of the new function.

Write for shortness

$$X = z_0 + Az_2 + Hz_3, \quad Y = z_1 + Hz_2 + Bz_3,$$

and suppose x changed into $x+1$: we have z_0, z_1, z_2, z_3 increased by a_0, a_1, a_2, a_3 respectively; and thence X, Y increased by $a_0 + Aa_2 + Ha_3, a_1 + Ha_2 + Ba_3$; the theta-function is thus changed into

$$\Theta(X + a_0 + Aa_2 + Ha_3, Y + a_1 + Ha_2 + Ba_3),$$

viz. the arguments are increased by multiples of the quarter-periods $(1, 0, A, H)$ and $(0, 1, H, B)$.

And, similarly, by the change of y into $y + 1$, the theta-function is changed into

$$\Theta(X + b_o + Ab_2 + Hb_3, Y + b_1 + Hb_2 + Bb_3),$$

or the arguments are increased by integer multiples of the quarter-periods.

47. But suppose next that x, y are changed into $x + A', y + H'$: we have z_o, z_1, z_2, z_3 increased by

$$A'a_o + H'b_o, \quad A'a_1 + H'b_1, \quad A'a_2 + H'b_2, \quad A'a_3 + H'b_3,$$

and thence

$$\begin{aligned} X \text{ increased by } A'(a_o + Aa_2 + Ha_3) + H'(b_o + Ab_2 + Hb_3) &= c_o + Ac_2 + Hc_3, \\ Y \quad, \quad A'(a_1 + Ha_2 + Ba_3) + H'(b_1 + Hb_2 + Bb_3) &= c_1 + Hc_2 + Bc_3, \end{aligned}$$

since these equalities are the before-mentioned equations

$$P(Ta_o + Pa_2 + Qa_3) + Q(Tb_o + Pb_2 + Qb_3) - T(Tc_o + Pc_2 + Qc_3) = 0,$$

$$P(Ta_1 + Qa_2 + Ra_3) + Q(Tb_1 + Qb_2 + Rb_3) - T(Tc_1 + Qc_2 + Rc_3) = 0.$$

It thus appears that, by the change of x, y into $x + A', y + H'$, the theta-function is changed into

$$\Theta(X + c_o + Ac_2 + Hc_3, Y + c_1 + Hc_2 + Bc_3),$$

viz. we have again the arguments increased by integer multiples of the quarter-periods; and in like manner, by the change of x, y into $x + H', y + B'$, the theta-function is changed into

$$\Theta(X + d_o + Ad_2 + Hd_3, Y + d_1 + Hd_2 + Bd_3),$$

or the arguments are again increased by integer multiples of the quarter-periods.

48. We have to complete the verifications, first for the equation

$$\Pi(x + 1, y) = (-)^\mu \Pi(x, y).$$

Reverting to the definition of Π , we have

$$\begin{aligned} (-)^\mu \Pi(x + 1, y) \div \Pi(x, y) &= (-)^{\mu+a_2+a_3+a_o a_2+a_1 a_3} \exp. -i\pi\{a_o a_2 + a_1 a_3\}; \\ \Theta(X + a_o + Aa_2 + Ha_3, Y + a_1 + Ha_2 + Ba_3) \div \Theta(X, Y) \\ &= \exp. i\pi[a_o z_2 + a_2 z_o + a_1 z_3 + a_3 z_1 + a_o a_2 + a_1 a_3] \times \\ &\quad \exp. i\pi(2(Aa_2 + Ha_3)z_2 + 2(Ha_2 + Ba_3)z_3 + (A, H, B \mid a_2, a_3)^2); \end{aligned}$$

the second line of this is

$$= (-)^{\mu+a_2+a_3+a_o a_2+a_1 a_3} \exp. i\pi\{-2a_2 X - 2a_3 Y - (A, H, B \mid a_2, a_3)^2\},$$

and the right-hand side of the equation will thus be $= 1$ if only the whole argument of the exponential be $=$; that is, omitting the terms which destroy each other, and the common factor $i\pi$, if only

$$\begin{aligned} &-2a_2 X - 2a_3 Y \\ &+ a_o z_2 + a_2 z_o + a_1 z_3 + a_3 z_1 \\ &+ 2(Aa_2 + Ha_3)z_2 + 2(Ha_2 + Ba_3)z_3 = 0. \end{aligned}$$

Substituting herein for X, Y their values

$$z_o + Az_2 + Hz_3, \quad z_1 + Hz_2 + Bz_3,$$

the equation becomes

$$-a_2 z_o - a_3 z_1 + a_1 z_3 + a_o z_2 = 0,$$

and finally substituting herein for z_o, z_1, z_2, z_3 their values, the coefficient of x is identically $= 0$, and the coefficient of y is

$$-a_2b_o - a_3b_1 + a_1b_3 + a_ob_2,$$

which is $(02 + 13)ab$, and is thus $= 0$. This completes the proof of the equation

$$\Pi(x + 1, y) = (-)^\mu \Pi(x, y);$$

and we have of course a precisely similar proof for the equation

$$\Pi(x, y + 1) = (-)^\nu \Pi(x, y).$$

49. For the next equation, writing for shortness $\alpha_o, \alpha_1, \alpha_2, \alpha_3$ for $A'a_o + H'b_o, A'a_1 + H'b_1, A'a_2 + H'b_2, A'a_3 + H'b_3$, so that z_o, z_1, z_2, z_3 are increased by $\alpha_o, \alpha_1, \alpha_2, \alpha_3$ respectively, we have

$$\begin{aligned} (-)^q \Pi(x + A', y + H') \div \Pi(x, y) &= (-)^{q+a_2+a_3+\alpha_o\alpha_2+\alpha_1\alpha_3} \exp. -i\pi\{\alpha_o\alpha_2 + \alpha_1\alpha_3\}; \\ \Theta(X + c_o + Ac_2 + Hc_3, Y + c_1 + Hc_2 + Bc_3) \div \Theta(X, Y) \\ &= \exp. i\pi\{\alpha_o z_2 + \alpha_2 z_o + \alpha_1 z_3 + \alpha_3 z_1 + \alpha_o\alpha_2 + \alpha_1\alpha_3\} \times \\ &\quad \exp. i\pi\{2(A\alpha_2 + H\alpha_3)z_2 + 2(H\alpha_2 + B\alpha_3)z_3 + (A, H, B \mid \alpha_2, \alpha_3)^2\}. \end{aligned}$$

The second line is here

$$= (-)^{q+a_2+a_3+\alpha_o\alpha_2+\alpha_1\alpha_3} \exp. i\pi\{-2c_2X - 2c_3Y - (A, H, B \mid c_2, c_3)^2\},$$

and the whole expression should be

$$= \exp. i\pi\{-2kx - kA'\}.$$

Hence bringing these terms over to the left-hand side, the equation will be satisfied if only the whole argument of the exponential be $=$; viz. omitting the factor $i\pi$, the equation will be satisfied if only

$$\begin{aligned} &\alpha_o z_2 + \alpha_2 z_o + \alpha_1 z_3 + \alpha_3 z_1 + \alpha_o\alpha_2 + \alpha_1\alpha_3 \\ &+ 2(A, H, B \mid \alpha_2, \alpha_3 \mid z_2, z_3) + (A, H, B \mid \alpha_2, \alpha_3)^2 \\ &- 2c_2X - 2c_3Y - (A, H, B \mid c_2, c_3)^2 + k(2x + A') = 0. \end{aligned}$$

Substituting here for X, Y their values

$$z_o + Az_2 + Hz_3, \quad z_1 + Hz_2 + Bz_3,$$

and attending to the values

$$a_o x + b_o y, \quad a_1 x + b_1 y, \quad a_2 x + b_2 y, \quad a_3 x + b_3 y$$

of z_o, z_1, z_2, z_3 , the equation contains a term in x , a term in y , and a constant term; and we may consider these terms separately.

50. The term in x will vanish if

$$\alpha_o a_2 + \alpha_2 a_o + \alpha_1 a_3 + \alpha_3 a_1 + 2(A\alpha_2 + H\alpha_3)a_2 + 2(H\alpha_2 + B\alpha_3)a_3 - 2c_2(a_o + Aa_2 + Ha_3) - 2c_3(a_1 + Ha_2 + Ba_3) + 2k = 0,$$

and substituting herein for $\alpha_o, \alpha_1, \alpha_2, \alpha_3$ their values

$$A'a_o + H'b_o, \quad A'a_1 + H'b_1, \quad A'a_2 + H'b_2, \quad A'a_3 + H'b_3,$$

the equation becomes

$$\begin{aligned} &A'[2a_o a_2 + 2a_1 a_3 + 2(Aa_2 + Ha_3)a_2 + 2(Ha_2 + Ba_3)a_3] \\ &+ H'[a_o b_2 + a_2 b_o + a_1 b_3 + a_3 b_1 + 2(Aa_2 + Ha_3)b_2 + 2(Ha_2 + Ba_3)b_3] \\ &- 2c_2(a_o + Aa_2 + Ha_3) - 2c_3(a_1 + Ha_2 + Ba_3) + 2k = 0, \end{aligned}$$

or observing that the first and second lines may be expressed in the form

$$\begin{aligned} & A'[2a_2(a_o + Aa_2 + Ha_3) + 2a_3(a_1 + Ha_2 + Ba_3)] \\ & + H'[2b_2(a_o + Aa_2 + Ha_3) + 2b_3(a_1 + Ha_2 + Ba_3)] - (a_ob_2 - a_2b_o + a_1b_3 - a_3b_1) \end{aligned}$$

where the term $a_ob_2 - a_2b_o + a_1b_3 - a_3b_1$, that is, $(02 + 13)ab$, is $= 0$, and may therefore be omitted, the whole equation, omitting the factor 2, which divides out, is

$$(A'a_2 + H'b_2 - c_2)(a_o + Aa_2 + Ha_3) + (A'a_3 + H'b_3 - c_3)(a_1 + Ha_2 + Ba_3) + k = 0,$$

an equation which, in the form

$$(-c_2T + a_2P + b_2Q)(a_oT + a_2P + a_3Q) + (-c_3T + a_3P + b_3Q)(a_1T + a_2Q + a_3R) + kTT = 0,$$

has been above shown to be true. The term in x thus vanishes.

51. The term in y will vanish if

$$\begin{aligned} & \alpha_ob_2 + \alpha_2b_o + \alpha_1b_3 + \alpha_3b_1 + 2(A\alpha_2 + H\alpha_3)b_2 + 2(H\alpha_2 + B\alpha_3)b_3 \\ & - 2c_2(b_o + Ab_2 + Hb_3) - 2c_3(b_1 + Hb_2 + Bb_3) = 0; \end{aligned}$$

and this is in a similar manner reduced to

$$(A'a_2 + H'b_2 - c_2)(b_o + Ab_2 + Hb_3) + (A'a_3 + H'b_3 - c_3)(b_1 + Hb_2 + Bb_3) = 0,$$

an equation which, in the form

$$(-c_2T + a_2P + b_2Q)(b_oT + b_2P + b_3Q) + (-c_3T + a_3P + b_3Q)(b_1T + b_2Q + b_3R) = 0,$$

has been above shown to be true. The term in y thus vanishes.

52. It only remains to show that the constant term also vanishes, viz. that we have

$$\alpha_o\alpha_2 + \alpha_1\alpha_3 + (A, H, B \mid \alpha_2, \alpha_3)^2 - c_oc_2 - c_1c_3 - (A, H, B \mid c_2, c_3)^2 + kA' = 0.$$

Substituting here for $\alpha_o, \alpha_1, \alpha_2, \alpha_3$ their values

$$= A'a_o + H'b_o, \quad A'a_1 + H'b_1, \quad A'a_2 + H'b_2, \quad A'a_3 + H'b_3,$$

the equation becomes

$$\begin{aligned} & A'^2(a_oa_2 + a_1a_3) + A'H'(a_ob_2 + a_2b_o + a_1b_3 + a_3b_1) + H'^2(b_ob_2 + b_1b_3) \\ & + A[A'^2a_2^2 + 2A'H'a_2b_2 + H'^2b_2^2] \\ & + 2H[A'^2a_2a_3 + A'H'(a_2b_3 + a_3b_2) + H'^2b_2b_3] \\ & + B[A'^2a_3^2 + 2A'H'a_3b_3 + H'^2b_3^2] \\ & - c_oc_2 - c_1c_3 - (A, H, B \mid c_2, c_3)^2 + kA' = 0. \end{aligned}$$

This may be written

$$\begin{aligned} & A'^2\{a_2(a_o + Aa_2 + Ha_3) + a_3(a_1 + Ha_2 + Ba_3)\} \\ & + A'H'\{b_2(a_o + Aa_2 + Ha_3) + b_3(a_1 + Ha_2 + Ba_3) \\ & \quad + a_2(b_o + Ab_2 + Hb_3) + a_3(b_1 + Hb_2 + Bb_3)\} \\ & + H'^2\{b_2(b_o + Ab_2 + Hb_3) + b_3(b_1 + Hb_2 + Bb_3)\} \\ & - c_2(c_o + Ac_2 + Hc_3) - c_3(c_1 + Hc_2 + Bc_3) + kA' = 0. \end{aligned}$$

Writing herein

$$\begin{aligned} c_o + Ac_2 + Hc_3 &= (a_o + Aa_2 + Ha_3)A' + (b_o + Ab_2 + Hb_3)H', \\ c_1 + Hc_2 + Bc_3 &= (a_1 + Ha_2 + Ba_3)A' + (b_1 + Hb_2 + Bb_3)H', \end{aligned}$$

equations which are true in virtue of

$$\Theta = 0,$$

the equation becomes

$$\begin{aligned} & (a_o + Aa_2 + Ha_3) A'(A'a_2 + H'b_2 - c_2) \\ & + (a_1 + Ha_2 + Ba_3) A'(A'a_3 + H'b_3 - c_3) \\ & + (b_o + Ab_2 + Hb_3) H'(A'a_2 + H'b_2 - c_2) \\ & + (b_1 + Hb_2 + Bb_3) H'(A'a_3 + H'b_3 - c_3) + kA' = 0, \end{aligned}$$

that is,

$$\begin{aligned} & A'[(a_o + Aa_2 + Ha_3)(-c_2 + A'a_2 + H'b_2) + (a_1 + Ha_2 + Ba_3)(-c_3 + A'a_3 + H'b_3) + k] \\ & + H'[(b_o + Ab_2 + Hb_3)(-c_2 + A'a_2 + H'b_2) + (b_1 + Hb_2 + Bb_3)(-c_3 + A'a_3 + H'b_3)] = 0; \end{aligned}$$

and we have the coefficients of A' and H' each = 0, in virtue of

$$\begin{aligned} & (a_oT + a_2P + a_3Q)(-c_2T + a_2P + b_2Q) + (a_1T + a_2Q + a_3R)(-c_3T + a_3P + b_3Q) + kTT = 0, \\ & (b_oT + b_2P + b_3Q)(-c_2T + a_2P + b_2Q) + (b_1T + b_2Q + b_3R)(-c_3T + a_3P + b_3Q) = 0. \end{aligned}$$

53. This completes the proof of the equation

$$\Pi(x + A', y + H') = (-)^q \Pi(x, y) \exp. - i\pi k(2x + A');$$

the proof of the remaining equation

$$\Pi(x + H', y + B') = (-)^r \Pi(x, y) \exp. - i\pi k(2y + B'),$$

is of course precisely similar.

It has already been seen that $\mu'q + \nu'r = \mu q + \nu r$; we have

$$\Theta(-X, -Y) = (-)^{\mu+\nu} \Theta(X, Y),$$

and thence

$$\Pi(-x, -y) = (-)^{\mu+\nu} \Pi(x, y),$$

that is,

$$\Pi(-x, -y) = (-)^{\mu'+\nu'} \Pi(x, y),$$

which is the last of the equations which should be satisfied by the function $\Pi(x, y)$.

RECAPITULATION, AND FINAL FORM. Art. Nos. 54 to 58.

54. Recapitulating, we have a Hermitian k -matrix (k an odd prime)

$$\begin{pmatrix} a_o, & a_1, & a_2, & a_3 \\ b_o, & b_1, & b_2, & b_3 \\ c_o, & c_1, & c_2, & c_3 \\ d_o, & d_1, & d_2, & d_3 \end{pmatrix},$$

susceptible of $(1 + k + k^2 + k^3)$ forms; further, writing for shortness

$$\begin{aligned} z_o &= a_o x + b_o y, \\ z_1 &= a_1 x + b_1 y, \\ z_2 &= a_2 x + b_2 y, \\ z_3 &= a_3 x + b_3 y, \end{aligned}$$

and then

$$X = z_o + Az_2 + Hz_3, \quad Y = z_1 + Hz_2 + Bz_3,$$

and assuming

$$\Pi(x, y) = \Theta\left(\begin{smallmatrix} \mu, & \nu \\ q, & r \end{smallmatrix}\right)(X, Y)(A, H, B) \exp . i\pi\{(z_o z_2 + z_1 z_3) + (A, H, B \mid z_2, z_3)^2\},$$

then $\Pi(x, y)$ satisfies the equations

$$\begin{aligned} \Pi(x+1, y) &= (-)^{\mu} \Pi(x, y), \\ \Pi(x, y+1) &= (-)^{\nu} \Pi(x, y), \\ \Pi(x+A', y+H') &= (-)^q \Pi(x, y) \exp . -i\pi k(2x+A'), \\ \Pi(x+H', y+B') &= (-)^r \Pi(x, y) \exp . -i\pi k(2y+B'), \\ \Pi(-x, -y) &= (-)^{\mu'+\nu'} \Pi(x, y), \end{aligned}$$

and it is consequently a function

$$\Pi\left(\begin{smallmatrix} \mu', & \nu' \\ q', & r' \end{smallmatrix}\right)(x, y)(A', H', B')(K, \text{def.}),$$

and as such, contains linearly $\frac{1}{2}(k^2 + 1)$ constants.

55. The values of the new parameters are given as above, viz. T', P', Q', R', S' being linear functions of T, P, Q, R, S (the coefficients in these relations being given functions of the coefficients $(a, b, c, d \mid 1)_{0,1,2,3}$ of the Hermitian matrix), then

$$1 : A : H : B : H^2 - AB = T : P : Q : R : S,$$

$$1 : A' : H' : B' : H'^2 - A'B' = T' : P' : Q' : R' : S',$$

which represent the required relations.

56. We consider four such functions $\Pi(x, y)$, derived from Θ -functions having respectively the characteristics

$$\left(\begin{smallmatrix} \mu_o, & \nu_o \\ q_o, & r_o \end{smallmatrix}\right), \left(\begin{smallmatrix} \mu_1, & \nu_1 \\ q_1, & r_1 \end{smallmatrix}\right), \left(\begin{smallmatrix} \mu_2, & \nu_2 \\ q_2, & r_2 \end{smallmatrix}\right), \left(\begin{smallmatrix} \mu_3, & \nu_3 \\ q_3, & r_3 \end{smallmatrix}\right),$$

being a (0123) system; and having consequently the characteristics

$$\left(\begin{smallmatrix} \mu'_o, & \nu'_o \\ q'_o, & r'_o \end{smallmatrix}\right), \left(\begin{smallmatrix} \mu'_1, & \nu'_1 \\ q'_1, & r'_1 \end{smallmatrix}\right), \left(\begin{smallmatrix} \mu'_2, & \nu'_2 \\ q'_2, & r'_2 \end{smallmatrix}\right), \left(\begin{smallmatrix} \mu'_3, & \nu'_3 \\ q'_3, & r'_3 \end{smallmatrix}\right),$$

which also form a (0123) system; say the four are $\Pi_o, \Pi_1, \Pi_2, \Pi_3$.

This being so, consider four theta-functions

$$\theta(x, y)(A', H', B')$$

having respectively the characteristics

$$\left(\begin{smallmatrix} \mu'_o, & \nu'_o \\ q'_o, & r'_o \end{smallmatrix}\right), \left(\begin{smallmatrix} \mu'_1, & \nu'_1 \\ q'_1, & r'_1 \end{smallmatrix}\right), \left(\begin{smallmatrix} \mu'_2, & \nu'_2 \\ q'_2, & r'_2 \end{smallmatrix}\right), \left(\begin{smallmatrix} \mu'_3, & \nu'_3 \\ q'_3, & r'_3 \end{smallmatrix}\right),$$

which as already mentioned form a (0123) system; form with these the four sums

$$\Sigma \theta_o^a \theta_1^b \theta_2^c \theta_3^d,$$

where in each case $a + b + c + d = k$, but in the first sum a , in the second sum b , in the third sum c , and in the fourth sum d is of contrary parity to the other three letters (even, if they are odd; odd, if they are even), say the four sums are $\Sigma_o, \Sigma_1, \Sigma_2, \Sigma_3$, respectively; each sum depends linearly on $\frac{1}{2}(k^2 + 1)$ constants.

The general transformation theorem then is

$$\Sigma_o = \Pi_o, \quad \Sigma_1 = \Pi_1, \quad \Sigma_2 = \Pi_2, \quad \Sigma_3 = \Pi_3;$$

each of which equations in fact represents $\frac{1}{2}(k^2 + 1)$ equations; viz. on the left-hand side we assign to the constants $\frac{1}{2}(k^2 + 1)$ systems of values at pleasure, then to each such system there corresponds on the right-hand side a determinate system of values of the $\frac{1}{2}(k^2 + 1)$ constants.

57. The results may be presented in a more symmetrical form. Writing ξ, η, ζ, ω for the foregoing x, y, z, w , we may consider

$$\Theta\left(\begin{matrix} \mu, & \nu \\ q, & r \end{matrix}\right)(\xi + A'\zeta + H'\omega, \eta + H'\zeta + B'\omega)(A', H', B')$$

as a function of the four arguments ξ, η, ζ, ω , and express it by

$$\Phi\left(\begin{matrix} \mu, & \nu \\ q, & r \end{matrix}\right)(\xi, \eta, \zeta, \omega)(A', H', B').$$

Writing then

$$\begin{aligned} x &= \xi + A'\zeta + H'\omega, \\ y &= \eta + H'\zeta + B'\omega, \\ X, Y, Z, W &= \begin{pmatrix} a_o & b_o & c_o & d_o \\ a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \\ a_3 & b_3 & c_3 & d_3 \end{pmatrix} (\xi, \eta, \zeta, \omega), \end{aligned}$$

we have

$$\begin{aligned} z_o &= a_o x + b_o y = a_o(\xi + A'\zeta + H'\omega) + b_o(\eta + H'\zeta + B'\omega) \\ &= (a_o \xi + b_o \eta) + A'(a_o \zeta + b_o \omega) + H'(a_o \omega + b_o \dots) \end{aligned}$$

and also

$$\begin{aligned} X + AZ + HW &= (a_o + Aa_2 + Ha_3) \xi \\ &\quad + (b_o + Ab_2 + Hb_3) \eta \\ &\quad + A(c_o + Ac_2 + Hc_3) \zeta \\ &\quad + H(c_o + \dots) \omega + (c_o + Ac_2 + Hc_3) \zeta + (d_o + Ad_2 + Hd_3) \omega, \end{aligned}$$

or since, as above,

$$\begin{aligned} A'(a_o + Aa_2 + Ha_3) + H'(b_o + Ab_2 + Hb_3) &= c_o + Ac_2 + Hc_3, \\ H'(a_o + Aa_2 + Ha_3) + B'(b_o + Ab_2 + Hb_3) &= d_o + Ad_2 + Hd_3, \end{aligned}$$

we find

$$z_o + Az_2 + Hz_3 = X + AZ + HW,$$

and in like manner another equation, viz. we have

$$z_1 + Hz_2 + Bz_3 = Y + HZ + BW.$$

Moreover

$$\begin{aligned} z_o z_2 + z_1 z_3 + (A, H, B \mid z_2, z_3)^2 &= z_2(z_o + Az_2 + Hz_3) + z_3(z_1 + Hz_2 + Bz_3) \\ &= (a_2 x + b_2 y)(X + AZ + HW) + (a_3 x + b_3 y)(Y + HZ + BW) \\ &= a_2(\xi + A'\zeta + H'\omega)(X + AZ + HW) + a_3(\xi + A'\zeta + H'\omega)(Y + HZ + BW) + \dots \end{aligned}$$

which qua function of X, Y, Z, W may be called χ .

58. And we then have the final result in the following form, viz.

$$\exp . i \pi \chi \cdot \Phi \left(\begin{matrix} \mu, & \nu \\ q, & r \end{matrix} \right) (X, Y, Z, W)(A, H, B),$$

in each of the four forms, is a homogeneous function of the order k of the corresponding four forms

$$\Phi \left(\begin{matrix} \mu', & \nu' \\ q', & r' \end{matrix} \right) (\xi, \eta, \zeta, \omega)(A', H', B').$$

849. On the Invariants of a Linear Differential Equation.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xxi. (1886), pp. 257–261.]

CONSIDER the equation of the second order

$$\frac{d^2 y}{dx^2} + 2p \frac{dy}{dx} + qy = 0;$$

if we effect a transformation of the dependent variable y , say $y = f \cdot Y$, where f is an arbitrary function of x , we obtain a new equation

$$\frac{d^2 Y}{dx^2} + 2P \frac{dY}{dx} + QY = 0,$$

where

$$P = p + \frac{f'}{f},$$

$$Q = q + 2p \frac{f'}{f} + \frac{f''}{f},$$

(the accents denoting differentiation in regard to x); and we thence establish the identity

$$Q - P^2 - P' = q - p^2 - p',$$

viz. we have $q - p^2 - p'$ a function possessing this invariantive property, but, as remarked by Mr Harley, it is the analogue rather of a seminvariant than of an invariant; I will call it an α -seminvariant of the differential equation. The class of function was considered long ago by Sir James Cockle, and more recently by Mr Malet; see Mr Harley's paper "Professor Malet's Classes of Invariants identified with Sir James Cockle's Criticoids," *Proc. R. Soc.*, vol. xxxviii. (1884), pp. 45–57.

Effecting for the same equation a transformation of the independent variable, say $x = \phi$, an arbitrary function of X , we obtain a new equation

$$\frac{d^2 y}{dX^2} + 2P \frac{dy}{dX} + Qy = 0,$$

where

$$P = p\phi_1 - \frac{1}{2} \frac{\phi_2}{\phi_1},$$

$$Q = q\phi_1^2,$$

(the subscript numbers denoting differentiation in regard to X). There is not in the present case any function possessing a like invariantive property, say there is not any β -seminvariant; but such functions exist for differential equations of the third and higher orders, and have been considered by Sir James Cockle and Mr Malet. It is to be noticed that, attempting to obtain a β -seminvariant, we are led to the equation

$$Q - P^2 - P_1 = \phi_1^2(q - p^2 - p') + \frac{1}{2} \left\{ \frac{\phi_3}{\phi_1} - \frac{3}{2} \left(\frac{\phi_2}{\phi_1} \right)^2 \right\},$$

which but for the last term would be invariantive. It is curious to find this last term presenting itself in the form of a Schwarzian derivative.

Passing to the differential equation of the third order

$$\frac{d^3y}{dx^3} + 3p \frac{d^2y}{dx^2} + 3q \frac{dy}{dx} + ry = 0,$$

here the transformation $y = f \cdot Y$ of the dependent variable gives the new equation

$$\frac{d^3Y}{dx^3} + 3P \frac{d^2Y}{dx^2} + 3Q \frac{dY}{dx} + RY = 0,$$

where

$$\begin{aligned} P &= p + \frac{f'}{f}, \\ Q &= q + 2p \frac{f'}{f} + \frac{f''}{f}, \\ R &= r + 3q \frac{f'}{f} + 3p \frac{f''}{f} + \frac{f'''}{f}, \end{aligned}$$

and we thence derive as well the before-mentioned α -seminvariant $q - p^2 - p'$, as also a new α -seminvariant $r - 3pq + 2p^3 - p''$.

Again, effecting the transformation $x = \phi$ of the independent variable, we have the new equation

$$\frac{d^3y}{dX^3} + 3P \frac{d^2y}{dX^2} + 3Q \frac{dy}{dX} + Ry = 0,$$

where

$$\begin{aligned} P &= p\phi_1 - \frac{\phi_2}{\phi_1}, \\ Q &= q\phi_1^2 - p\phi_2 + \left(\frac{\phi_2}{\phi_1}\right)^2 - \frac{1}{3} \frac{\phi_3}{\phi_1}, \\ R &= r\phi_1^3. \end{aligned}$$

We thence obtain the identities

$$\begin{aligned} \frac{P_1 + 2P^2 - 3Q}{R^{\frac{2}{3}}} &= \frac{p' + 2p^2 - 3q}{r^{\frac{2}{3}}}, \\ \frac{R_1 + 3PR}{R^{\frac{4}{3}}} &= \frac{r' + 3pr}{r^{\frac{4}{3}}}; \end{aligned}$$

so that we have $\frac{p' + 2p^2 - 3q}{r^{\frac{2}{3}}}$ and $\frac{r' + 3pr}{r^{\frac{4}{3}}}$ as β -seminvariants of the differential equation of the third order.

But these are by no means the best conclusions; it is shown by M. Halphen in his great Memoir, “Mémoire sur la réduction des équations différentielles linéaires aux formes intégrables,” Sav. Étrang., t. xxviii. (1884), pp. 1–297, see p. 127, that there is a function invariantive in regard to each of the two transformations, and which is thus an invariant, viz. this is the function

$$p'' - 3(q' - 2pp') + 2(r - 3pq + 2p^3);$$

this is for the first transformation unaltered, viz. it is

$$= P'' - 3(Q' - 2PP') + 2(R - 3PQ + 2P^3),$$

and for the second transformation it is only altered by the factor ϕ_1^{-3} , viz. it is

$$= \phi_1^{-3} \{P'' - 3(Q_1 - 2PP_1) + 2(R - 3PQ + 2P^3)\}.$$

It is interesting to directly verify this last result. Performing the differentiations in regard to X , we find without difficulty

$$\begin{aligned} P_1 &= \cdots, \\ Q_1 - 2PP_1 &= (q' - 2pp')\phi_1^2 - \cdots, \\ R - 3PQ + 2Q^3 &= (r - 3pq + 2p^3)\phi_1^3, \end{aligned}$$

and thence

$$P'' - 3(Q_1 - 2PP_1) + 2(R - 3PQ + 2Q^3) = \phi_1^3 \{p'' - 3(q' - 2pp') + 2(r - 3pq + 2p^3)\} \phi_1^{-2}.$$

But, introducing herein the derived functions in regard to x , we have

$$p_1 = p'\phi_1, \quad q_1 = q'\phi_1, \quad p_2 = p'\phi_1 + p''\phi_1^2,$$

whence

$$p_2\phi_1 - p_1\phi_2 = p''' \phi_1^3,$$

and the right-hand side becomes

$$= \phi_1^{-3} \{p'' - 3(q' - 2pp') + 2(r - 3pq + 2p^3)\},$$

which is the required result.

We have thus

$$p'' - 3(q' - 2pp') + 2(r - 3pq + 2p^3)$$

as an “invariant” of the differential equation

$$\frac{d^3y}{dx^3} + 3p \frac{d^2y}{dx^2} + 3q \frac{dy}{dx} + ry = 0.$$

It is to be remarked that this is not what M. Halphen calls a “differential invariant;” he uses this expression not in regard to a differential equation, but in regard to a curve defined by an equation between the coordinates (x, y) , and the differential invariant is a function of the derivatives $y', y'', y''', \dots$ which is either $= 0$ in virtue of the equation of the curve, or else, being put $= 0$, it determines certain singularities of the curve. Thus y'' is a differential invariant; the equation $y'' = 0$ determines the points of inflexion. Again

$$\frac{1}{3} \left(\frac{y''''}{y''} \right) - \frac{5}{3} \frac{y'''}{y''} \left(\frac{y''''}{y''} \right) + \frac{40}{9} \left(\frac{y'''}{y''} \right)^2$$

is a differential invariant, vanishing identically if the variables (x, y) are connected by any quadric equation whatever; it is thus the differential invariant of a conic.

This last differential invariant is intimately connected with the above-mentioned invariant

$$p'' - 3(q' - 2pp') + 2(r - 3pq + 2p^3),$$

viz. writing with M. Halphen

$$p = -\frac{1}{3} \frac{y'''}{y''}, \quad q = 0, \quad r = 0,$$

we have

$$p'' - 3(q' - 2pp') + 2(r - 3pq + 2p^3) = p'' + 6pp' + 4p^3.$$

It is moreover noticed by him that, writing

$$y'', y''', y'''', y''''' = 2a, 6b, 24c, 120d,$$

respectively, the function in $\{ \}$ becomes

$$\frac{20}{a^3} (a^2d - 3abc + 2b^3);$$

the form under which he had previously obtained the differential invariant of the conic. As remarked by Sylvester, it is mentioned pp. 19 and 20 in Boole's *Differential Equations* (Cambridge, 1859), that the general differential equation of a conic was obtained by Monge in the form

$$9y'^2y''' - 45y'y''y'''' + 40(y'')^3 = 0,$$

which (representing the differential coefficients as just mentioned) becomes $a^2d - 3abc + 2b^3 = 0$, but putting them $= a, b, c, d$ respectively, it becomes $9a^2d - 45abc + 40b^3 = 0$; the last-mentioned form presented itself to Sylvester in his theory of Reciprocants.

850. On Linear Differential Equations.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. *xxi.* (1886), pp. 321–331.]

1. THE researches of Fuchs, Thomé, Frobenius, Tannery, Floquet, and others, relate to linear differential equations of the form

$$p_0 \frac{d^m y}{dx^m} + p_1 \frac{d^{m-1} y}{dx^{m-1}} + \cdots + p_m y = 0,$$

or say

$$(p_0, p_1, \dots, p_m \mid d/dx, 1)^m y = 0,$$

where $p_0, p_1, \dots, p_m$ are rational and integral functions of the independent variable x ; any common factor of all the functions could of course be thrown out, and it is therefore assumed that the functions have no common factor. It is to be throughout understood that x and y denote complex magnitudes, which may be regarded as points in the infinite plane; viz. $x = \xi + i\eta$, is the point the coordinates whereof are ξ, η ; and similarly for y .

2. Suppose $x - a$ is not a factor of p_0 , the point $x = a$ is in this case said to be an ordinary point in regard to the differential equation; and let $y_0, y_1, y_2, \dots, y_{m-1}$ be arbitrary constants denoting the values of $y, \frac{dy}{dx}, \frac{d^2 y}{dx^2}, \dots, \frac{d^{m-1} y}{dx^{m-1}}$ for the point $x = a$. We can from the differential equation, and the equations derived therefrom by successive differentiations in regard to x , obtain the values for $x = a$ of the subsequent differential coefficients $\frac{d^m y}{dx^m}, \frac{d^{m+1} y}{dx^{m+1}}, \dots$, say these are $y_m, y_{m+1}, \dots$; viz. the value of each of these quantities will be determined as a linear function of $y_0, y_1, \dots, y_{m-1}$, and we thus have a development of y in positive integer powers of $x - a$, viz. this is

$$y = y_0 + y_1(x - a) + \frac{1}{1 \cdot 2} y_2(x - a)^2 + \cdots,$$

which will in fact be a sum of m determinate series multiplied by $y_0, y_1, \dots, y_{m-1}$ respectively; say the form is

$$y = y_0 X_0 + y_1 X_1 + \cdots + y_{m-1} X_{m-1},$$

where $X_0, X_1, \dots, X_{m-1}$ are each of them a series of positive integer powers of $x - a$. Each of these series will be convergent for values of x sufficiently near to a , or say for points x within the domain of the point a ; and since $y_0, y_1, \dots, y_{m-1}$ are arbitrary, each series separately will satisfy the differential equation; and we have thus m particular integrals of the differential equation.

3. Suppose next that $x - a$ is a factor of p_0 , the point $x = a$ is in this case said to be a singular point in regard to the differential equation. The foregoing process of development fails, as leading to infinite values of $\frac{d^m y}{dx^m}, \frac{d^{m+1} y}{dx^{m+1}}, \dots$; and we have to consider the developments of y which belong to the neighbourhood of the point $x = a$, or say to the domain of the singular point $x = a$. This has to be done separately in regard to each of the singular points, and among these we have, it may be, the point ∞ . To decide whether ∞ is or is not a singular point, we may in the differential equation write $x = 1/t$, and then transforming to this new variable, and throwing out any common factor, the coefficients of the transformed equation will be rational and integral functions of t , without any common factor; say these are $P_0, P_1, \dots, P_m$; if t is

not a factor of P_o , then ∞ will be an ordinary point of the original equation; but if t is a factor of P_o , then ∞ will be a singular point of the original equation.

4. In considering the singular point $x = a$, we may, it is clear, transform the equation to this point as origin, and it is convenient to do this; supposing it done, we have an equation wherein $x = 0$ is a singular point, viz. p_o contains x as a factor, and we have to consider the developments of y which belong to the domain of this singular point $x = 0$.

It is convenient to change the form of the differential equation by dividing the whole equation by the first coefficient p_o , and then expanding each of the quotients $\frac{p_i}{p_o}$ in a series of ascending powers of x . The new form is

$$\frac{d^m y}{dx^m} + p_1 \frac{d^{m-1} y}{dx^{m-1}} + \cdots + p_m y = 0,$$

or say

$$(1, p_1, p_2, \dots, p_m \mid d/dx, 1)^m y = 0,$$

where $p_1, p_2, \dots, p_m$ now denote each of them a series (finite or infinite) of integer powers of x , but containing only a finite number of negative powers of x .

5. Such an equation frequently admits of a “regular integral” of the form

$$y = x^\rho E(x),$$

where $E(x)$ is a series of positive integer powers $E_o + E_1 x + E_2 x^2 + \cdots$, (E_o not $= 0$, for this would imply a different value of ρ).¹ To determine whether this is so, we substitute in the differential equation for y the value in question $x^\rho E(x)$, thus obtaining a series

$$\Omega_o x^{\rho-\theta} + \Omega_1 x^{\rho-\theta+1} + \cdots,$$

(where θ is a determinate positive integer depending on the negative powers of x in the equation); the coefficients $\Omega_o, \Omega_1, \dots$ are functions of ρ of an order not exceeding m , and contain also the coefficients $E_o, E_1, E_2, \dots$ linearly; in particular, Ω_o contains E_o as a factor, say its value is $= E_o \Pi$. The series should vanish identically. Supposing that Π contains ρ , then we have $\Pi = 0$, an equation of an order not exceeding m for the determination of ρ . For any root $\rho = \rho_o$ of this equation, E_o remains arbitrary and may be taken $= 1$; the equations $\Omega_1 = 0, \Omega_2 = 0, \dots$ then serve to determine the ratios of the remaining coefficients $E_1, E_2, \dots$ to E_o ; and we thus have the solution $y = x^{\rho_o} (1 + E_1 x + E_2 x^2 + \cdots)$, where ρ_o and the coefficients have determinate values.

6. I stop to notice a curious form of illusory solution; the assumed form of solution is

$$y = x^\rho (\cdots + E_{-2} x^{-2} + E_{-1} x^{-1} + E_o + E_1 x + \cdots),$$

the series being a double series extending both ways to infinity, or say a back-and-forward series; we have here a series of equations

$$\dots, \Omega_{-1} = 0, \Omega_o = 0, \Omega_1 = 0, \dots,$$

which leave ρ undetermined, but determine the ratios of the several coefficients to one of these coefficients, say E_o ; or taking this $= 1$, we have a solution

$$y = x^\rho (\cdots + E_{-2} x^{-2} + E_{-1} x^{-1} + 1 + E_1 x + E_2 x^2 + \cdots),$$

where the coefficients are determinate functions of the arbitrary symbol ρ . Such a series is in general divergent for all values of the variable, and thus is altogether without meaning. As a simple instance, take the differential equation

$$\frac{dy}{dx} - y = 0,$$

which is satisfied by

$$y = \{ \cdots + (\rho - 1) \rho x^{\rho-2} + \rho x^{\rho-1} + x^\rho + \frac{x^{\rho+1}}{\rho + 1} + \cdots \};$$

¹The expression regular integral is afterwards used in a more general sense, see post, No. 10.

see my paper, Cayley, Note on Riemann's paper, "Versuch einer allgemeinen Auffassung der Integration und Differentiation," Werke, pp. 331–344 (Math. Ann. t. xvi. (1880), pp. 81, 82), [751].

7. A more general form of integral is Thomé's "normal elementary integral,"

$$y = e^w x^\rho E(x)$$

where w is $= \frac{c_\alpha}{x^\alpha} + \cdots + \frac{c_1}{x}$ a finite series of negative powers of x (α a positive integer, $= 2$ at least).

To discover whether such a form exists, observe that, writing for a moment $\frac{y'}{y} = z$ and so for the other symbols, we have

$$z = w' + \frac{\rho}{x} + \frac{E'(x)}{E(x)},$$

where the last term $\frac{E'(x)}{E(x)}$ may be expanded as a series of positive integer powers of x , so that, writing $z = \frac{y'}{y}$, we have

$$z = \cdots - \frac{\alpha c_\alpha}{x^{\alpha+1}} - \cdots - \frac{c_1}{x^2} + \frac{\rho}{x} + l_0 + l_1 x + \cdots,$$

we can, by introducing into the differential equation the new variable $z (= \frac{y'}{y})$ in place of y , obtain for z a differential equation (not a linear one) of the order $m - 1$, and this should be satisfied by the series in question, viz. a series containing a finite number of negative powers of x ; we endeavour to determine the first term $\frac{-(\alpha - 1)c_{\alpha-1}}{x^\alpha}$ or say $\frac{A}{x^\alpha}$, where the exponent α is a positive integer which is to be determined, and A is not $= 0$; this is done by a well-known process; we write in the equation $z = \frac{A}{x^\alpha}$, and (if possible to do so) determine α (α a positive integer $= 2$ at least) by the condition that two or more of the terms shall have the same negative index $-p$ preceding in order all the other indices, viz. the absolute value of p must be greater than the absolute value of any other negative index; we then equate to zero the whole coefficient of the term in question x^{-p} , and thus obtain a value not $= 0$ of A . And having thus obtained the first term $\frac{A}{x^\alpha}$, we can, by assuming the form

$$z = \frac{A}{x^\alpha} + \frac{B}{x^{\alpha-1}} + \cdots + \frac{K}{x^2} + \frac{L}{x} + \cdots,$$

with indeterminate coefficients, and substituting in the equation, find the remaining coefficients $B, \dots, K, L$; we then have

$$\frac{c_\alpha}{x^\alpha} + \cdots + \frac{c_1}{x} = w,$$

giving the value $\frac{c_\alpha}{x^\alpha} + \cdots + \frac{c_1}{x}$ of w ; instead of determining the subsequent coefficients from the z -equation, we may, from the original equation, writing therein $y = e^w Y$, obtain an equation for $Y (= e^{-w} y)$ which will be linear of the order m , and will admit of a regular integral $Y = x^\rho E(x)$. Or we may, going on a step further with the z -equation, find therefrom the coefficient L which is $= \rho$.

8. As an example, take the equation

$$\frac{d^2 y}{dx^2} + \left(\frac{3}{x} - \frac{1}{x^2} \right) \frac{dy}{dx} + \frac{1}{x^2} y = 0,$$

which has the elementary normal integral

$$y = e^{-\frac{1}{x}} \cdot \frac{1}{x}.$$

To investigate this, writing $\frac{y'}{y} = z$, we have $\frac{y''}{y} = z' + z^2$, and thence the z -equation

$$\frac{dz}{dx} + z^2 + \left(\frac{3}{x} - \frac{1}{x^2} \right) z + \frac{1}{x^2} = 0.$$

Substituting herein the value $z = \frac{A}{x^\alpha}$, we obtain the function

$$-\frac{A\alpha}{x^{\alpha+1}} + \frac{A^2}{x^{2\alpha}} - \frac{A}{x^{\alpha+1}} + \frac{1}{x^2},$$

and here the value $\alpha = 2$ gives the two terms $\frac{A^2}{x^4}$ and $-\frac{A}{x^4}$, each with a negative index -4 , the absolute value whereof is greater than the absolute values of the other indices; we then have $A^2 - A = 0$, giving a value $A = 1$, which is not $= 0$; and we have thus a first term $z = -\frac{1}{x^2}$, the complete value is, as it happens, the finite series $z = \frac{1}{x^2} - \frac{1}{x}$; and z being $= \frac{y'}{y}$, we thence obtain the integral in question, $y = e^{-\frac{1}{x}} \cdot \frac{1}{x}$.

9. Returning to the question of the regular integrals, we may consider the function

$$P(x^\rho) = (1, p_1, \dots, p_m \mid d/dx, 1)^m x^\rho;$$

say this is the so-called “determinirende Function,” but I will call it the Indicial function; this is a function of (x, ρ) ; we have therein terms arising from

$$\left(\frac{d}{dx}\right)^m x^\rho, \quad \left(\frac{d}{dx}\right)^{m-1} x^\rho, \dots,$$

viz. these are equal to $[\rho]^m x^{\rho-m}$, $[\rho]^{m-1} x^{\rho-m+1}$, ... respectively, but these will be multiplied by powers of x contained in the coefficients $p_1, p_2, \dots, p_m$ respectively; the function is thus of the degree m as regards ρ , but the coefficient of any power of x is of the degree m , or of some inferior degree, according to the terms $[\rho]^m$, $[\rho]^{m-1}$, ... which enter into the coefficient. The coefficient of the lowest power of x in the indicial function may be termed the indicial coefficient, or simply the indicial; and the equation obtained by equating this coefficient to zero the indicial equation. By what precedes, the indicial is a function of ρ , which is of the degree m at most, but which may be of any inferior degree, or even be an absolute constant. Hence, considering a regular integral $x^\rho E(x)$, the index ρ is determined by the indicial equation, and to each root of this equation there corresponds in general a regular integral $x^\rho E(x)$; the number of regular integrals is thus at most $= m$.

10. Suppose, first, that the indicial equation is of the degree m , and that the roots of this equation are all unequal; there are, in this case, m values, say $\rho_1, \rho_2, \dots, \rho_m$ of ρ , and each of these gives rise to a regular integral $x^\rho E(x)$, viz. there will be corresponding to each value of ρ a series of positive integer powers $E(x)$, which will be a convergent series for sufficiently small values of x . Some of these series may be finite series, viz. this will be the case if the indicial equation has roots differing from each other by integer values (but a finite series may of course be regarded as convergent); and in the case of two or more equal roots, it is necessary to extend the notion of a regular integral by including under it series multiplied into positive integer powers of $\log x$; but, with this modification, the conclusion holds good; if the indicial equation be of the degree m , the number of regular integrals will always be $= m$, viz. there will be m integrals involving series $E(x)$, or, it may be, $E(x)(\log x)^k$, where each series $E(x)$ is a convergent series for sufficiently small values of x .

11. But suppose the indicial equation is of an order $m - \gamma$ inferior to m , and to fix the ideas suppose that the roots are all unequal. There are in this case $m - \gamma$ values, say $\rho_1, \rho_2, \dots, \rho_{m-\gamma}$ of ρ , and each of these will apparently give rise to a regular integral $x^\rho E(x)$; but it may be that in any such case the series $E(x)$ is a series divergent for all values, however small, of the variable x , and that we thus have in appearance only, but not in reality, a regular integral; or say the integral is illusory. The conclusion is that the indicial equation being of a degree $m - \gamma$ inferior to m , the number of regular integrals is at most $= m - \gamma$; but it may have any less value than $m - \gamma$, or even there may be no regular integral. It is hardly necessary to remark that, if the indicial be an absolute constant, or say if the degree of the indicial equation be $= 0$, then there is no value of ρ , and consequently no regular integral.

12. By way of illustration, consider first the differential equation

$$\frac{d^2 y}{dx^2} + \frac{1}{x} \frac{dy}{dx} + \frac{1}{x^2} y = 0,$$

for which the degree of the indicial equation is equal to the order of the equation ($= 2$); viz. the indicial function is

$$= \rho(\rho - 1)x^{\rho-2} + \rho x^{\rho-2} + x^{\rho-2},$$

and the indicial equation is thus $\rho(\rho - 1) + \rho = 0$, that is, $\rho^2 = 0$, viz. there are here two roots, each $= 0$.

It is convenient not in the first instance to write $\rho = 0$, but to substitute in the equation the value

$$y = Ax^\rho + Bx^{\rho+1} + Cx^{\rho+2} + \dots$$

We thus find

$$0 = \begin{array}{ccc} \rho \cdot \rho - 1 \cdot A, & \overline{\rho + 1} \cdot \rho \cdot B, & \overline{\rho + 2} \cdot \overline{\rho + 1} \cdot C \\ + \rho A, & + \overline{\rho + 1} \cdot B, & + \overline{\rho + 2} \cdot C \\ + A, & + B, & + C, \end{array}$$

viz. we have the equations

$$\begin{aligned} \rho^2 A &= 0, \\ (\rho + 1)^2 B + A &= 0, \\ (\rho + 2)^2 C + B &= 0, \end{aligned}$$

where observe that in each equation the function of ρ , which multiplies the posterior coefficient, is of a degree superior to that which multiplies the other coefficient. The first equation gives $\rho = 0$; A arbitrary, but say $A = 1$; the values of $B, C, \dots$ are then

$$-\frac{1}{(\rho + 1)^2}, \quad +\frac{1}{(\rho + 1)^2(\rho + 2)^2}, \quad -\frac{1}{(\rho + 1)^2(\rho + 2)^2(\rho + 3)^2}, \quad \dots,$$

giving rise to a series which in this form, and when ρ is put equal to its value, $\rho = 0$, is at once seen to be convergent (even for indefinitely large values of x , but this is an accident; it would have been enough if the series had been convergent for sufficiently small values), viz. the series is

$$y = 1 - \frac{x}{1^2} + \frac{x^2}{1^2 \cdot 2^2} - \frac{x^3}{1^2 \cdot 2^2 \cdot 3^2} + \dots$$

There is another integral

$$y = \left(1 - \frac{x}{1^2} + \frac{x^2}{1^2 \cdot 2^2} - \dots\right) \log x + 2x - \frac{3}{4}x^2 + \frac{11}{108}x^3 - \frac{1}{1728}x^4 + \dots,$$

involving $\log x$ and a second series which is also convergent.

13. But consider next the before-mentioned equation

$$\frac{d^2 y}{dx^2} + \left(\frac{3}{x} - \frac{1}{x^2}\right) \frac{dy}{dx} + \frac{1}{x^2} y = 0,$$

for which the degree of the indicial equation is equal to order -1 ($= 1$); in fact, here the indicial function is

$$= \rho(\rho - 1)x^{\rho-2} + 3\rho x^{\rho-2} - \rho x^{\rho-3} + x^{\rho-2},$$

and the indicial is thus $= -\rho$, giving the index $\rho = 0$. There is thus at most one regular integral $E(x)$, but this is in fact an illusory one. Substituting in the equation the value

$$y = Ax^\rho + Bx^{\rho+1} + Cx^{\rho+2} + \dots,$$

we find

$$0 = \begin{array}{cccc} \rho(\rho - 1)A, & (\rho + 1)\rho B, & (\rho + 2)(\rho + 1)C, & \dots \\ -\rho A, & -(\rho + 1)B, & -(\rho + 2)C, & \dots \\ +3\rho A, & +3(\rho + 1)B, & +3(\rho + 2)C, & \dots \\ +A, & +B, & +C, & \end{array}$$

and we thus have the equations

$$\begin{aligned}\rho A &= 0, \\ (\rho + 1)^2 A - (\rho + 1) B &= 0, \\ (\rho + 2)^2 B - (\rho + 2) C &= 0, \\ (\rho + 3)^2 C - (\rho + 3) D &= 0,\end{aligned}$$

where observe that in each equation the function of ρ , which multiplies the posterior coefficient, is of a degree inferior to that which multiplies the other coefficient. The first equation then gives $\rho = 0$, A arbitrary, or say $= 1$; the other coefficients $B, C, \dots$ are then found to be

$$\rho + 1, \quad (\rho + 1)(\rho + 2), \quad (\rho + 1)(\rho + 2)(\rho + 3), \quad \dots,$$

or substituting for ρ its value $= 0$, the series is

$$1 + 1 \cdot x + 1 \cdot 2 \cdot x^2 + 1 \cdot 2 \cdot 3 \cdot x^3 + \dots,$$

which is divergent for any value whatever, however small, of x ; or the integral is as mentioned an illusory one.

14. The last-mentioned differential equation is reducible, viz. we have

$$\frac{d^2}{dx^2} + \left(\frac{3}{x} - \frac{1}{x^2} \right) \frac{d}{dx} + \frac{1}{x^2} = \left(\frac{d}{dx} + \frac{2}{x} \right) \left(\frac{d}{dx} + \frac{1}{x} - \frac{1}{x^2} \right), \text{ say } = QD,$$

where of course the order of the factors Q, D is material.

It is a general theorem that the indicial, belonging to the product P , is of a degree equal to the sum of the degrees of the indicials, belonging to the factors Q, D respectively; and, in fact, the indicial of P is (as was seen) of the degree 1; and the indicials of Q, D are of the degrees 1, 0 respectively. But, moreover, if the indicial of P is of a degree $m - \gamma$, less than m the order of the equation (in the present case $\gamma = 1$, $m = 2$, $m - \gamma = 1$), then, in order that the equation may have $m - \gamma$ regular integrals, it is a necessary and sufficient condition that P shall be decomposable into a product QD , where the factors Q, D (being functions of the same form as P) are of the orders γ and $m - \gamma$ respectively, and where, moreover, the second factor D shall have an indicial of the degree $m - \gamma$, and consequently the first factor Q an indicial of the degree zero. In the present case, it is the second factor which has an indicial of degree zero; and thus the condition is not satisfied. And accordingly the equation ought not to have $(m - \gamma) = 1$ regular integral; and we have seen that there is in fact no regular integral. The investigation serves to show in what sense it is that there is no regular integral, or generally in what sense it is that the number of regular integrals may be less than $m - \gamma$.

15. The theorem may, it appears to me, be stated in the more general form: the necessary and sufficient condition that the differential equation $P(y) = 0$ of the order m , but having an indicial of the degree $m - \gamma$, shall have $m - \gamma - \delta$ regular integrals is that the function P shall be a product of the form $P = QMD$, where the orders of Q, M, D are $\gamma + \delta - \theta, \theta, m - \gamma - \delta$ respectively, and the degrees of their indicials $\delta, 0, m - \gamma - \delta$ respectively; or, to denote this in a compendious manner, say

$$\underset{m-\gamma}{P}^m = \underset{\delta}{Q}^{\gamma+\delta-\theta} \underset{0}{M}^{\theta} \underset{m-\gamma-\delta}{D}^{m-\gamma-\delta}.$$

θ is an integer which is not $= 0$, and which is at most $= \gamma$, for otherwise the function Q of the order $\gamma + \delta - \theta$ could not have an indicial of the degree δ . If there is no regular integral, then $m = \gamma + \delta$, D is a mere constant or say unity, and the formula is

$$\underset{m-\gamma}{P}^m = \underset{m-\gamma}{Q}^{m-\theta} \underset{0}{M}^{\theta},$$

where θ is not $= 0$, and is at most $= m - \gamma$.

16. For differential equations of the form $P(y) = 0$ above considered, Floquet gave a theorem, which as afterwards remarked by him is not generally true, viz. this was $P = ABC \dots$, a product of linear factors

of the form $\frac{d}{dx} + \sum_{-i}^{\infty} C_i x^i$ (i an integer). The question of decomposition is considered in my next following paper “On Linear Differential Equations (the Theory of Decomposition),” [851], and by means of the formulæ there given (and by the process indicated ante No. 7), it could be decided whether any particular differential equation admits of a decomposition $P = ABC \cdots$, where the linear factors are functions not of the form just referred to, but of the form $\frac{d}{dx} + \sum C_i x^i$, in which the series contains only a finite number of negative powers, say this is a decomposition into linear regular factors.

851. On Linear Differential Equations (the Theory of Decomposition).

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xxi. (1886), pp. 331–335.]

1. IN the theory of linear differential equations the question arises, to decompose a quantic $\left(* \mid \frac{d}{dx}, 1\right)^n$ into linear factors: we have, for instance, a differential equation

$$\frac{d^2 y}{dx^2} + p \frac{dy}{dx} + qy = 0,$$

which is to be expressed in the form

$$\left(\frac{d}{dx} + \alpha\right)\left(\frac{d}{dx} + \beta\right)y = 0,$$

or say we have to express $\frac{d^2}{dx^2} + p \frac{d}{dx} + q$ as a product of linear factors

$$\left(\frac{d}{dx} + \alpha\right)\left(\frac{d}{dx} + \beta\right).$$

The problem is analogous to, but wholly distinct from, that of the resolution of an algebraic equation: using accents to denote differentiation in regard to x , the relation (in the simple case just referred to) between the coefficients (p, q) and the roots (α, β) is (not $p = \alpha + \beta$, $q = \alpha\beta$, but in place thereof) $p = \alpha + \beta$, $q = \alpha\beta + \beta'$; and it thus appears that there is the important distinction that, in the present problem, the order of the factors is not indifferent.

2. The problem may be solved when the general solution of the differential equation is known, or, what is the same thing, by means of n particular solutions of the equation; it is not here considered in this point of view, but the intention is to treat it directly by means of the relations between the coefficients and the roots: thus in the above case, $p = \alpha + \beta$, $q = \alpha\beta + \beta'$; we may (1) find first β and then α , viz. eliminating α we have $\beta^2 + \beta'(p - \beta) - q = 0$, β determined by a differential equation (not linear) of the first order; and then $\alpha = p - \beta$; or we may (2) find first α and then β , viz. eliminating β , we have $\alpha^2 - p\alpha + \alpha'(\alpha - p) + q = 0$, α determined by a differential equation (not linear) of the first order, and then $\beta = p - \alpha$.

In the case of a product

$$\left(\frac{d}{dx} + \alpha\right)\left(\frac{d}{dx} + \beta\right)\left(\frac{d}{dx} + \gamma\right) \cdots$$

of more than two factors, the roots might be determined in any order; but the two orders which naturally present themselves and which will be alone considered are, say, the reverse order $(\dots, \gamma, \beta, \alpha)$ and the direct order $(\alpha, \beta, \gamma, \dots)$.

3. *The Reverse Order.* Writing D for $\frac{d}{dx}$, we assume

$$\begin{aligned} D + \alpha &= (1, p_1 \mid D, 1), \\ (D + \alpha)(D + \beta) &= (1, p_2, q_2 \mid D, 1)^2, \\ (D + \alpha)(D + \beta)(D + \gamma) &= (1, p_3, q_3, r_3 \mid D, 1)^3, \end{aligned}$$

and so on. Then using accents to denote differentiation, we find

$$\begin{aligned}
 (1) \quad & p_1 = \alpha. \\
 (2) \quad & p_2 = \beta + p_1, \\
 & q_2 = \beta' + p_1\beta. \\
 (3) \quad & p_3 = \gamma + p_2, \\
 & q_3 = 2\gamma' + p_2\gamma + q_2, \\
 & r_3 = \gamma'' + p_2\gamma' + q_2\gamma. \\
 (4) \quad & p_4 = \delta + p_3, \\
 & q_4 = 3\delta' + p_3\delta + q_3, \\
 & r_4 = 3\delta'' + 2p_3\delta' + q_3\delta + r_3, \\
 & s_4 = \delta''' + p_3\delta'' + q_3\delta' + r_3\delta,
 \end{aligned}$$

where the law is obvious: thus in the last set of equations, the several columns (after the first) contain the factors p_3, q_3, r_3, s_3 respectively; and there is in each column a head term with the coefficient unity: omitting these head terms, we have in the several columns the sets $(\delta, 3\delta', 3\delta'', \delta''')$, $(\delta, 2\delta', \delta'')$, (δ, δ') , δ , of derivatives of the root δ .

4. We may from each set of equations determine in order the coefficients of lower rank contained in the equations, and the last equation of each set then gives an equation independent of these coefficients of lower rank. Thus set (2) gives

$$p_1 = -\beta + p_2; \quad 0 = -\beta' + p_2\beta - p_2 \cdot \beta.$$

Set (3) gives

$$\begin{aligned}
 p_2 &= -\gamma + p_3, & q_2 &= -2\gamma' + (\gamma + \gamma')p_3 + q_3, \\
 0 &= \gamma'' - \gamma \cdot 2\gamma' + p_3(\gamma\gamma' - \gamma'^2) + q_3(\gamma\gamma' - \gamma'\gamma).
 \end{aligned}$$

Set (4) gives

$$\begin{aligned}
 p_3 &= -\delta + p_4, \\
 q_3 &= -3\delta' + 5\delta\delta' - \delta'^2, \\
 0 &= \delta''' - 4\delta\delta'' + 6\delta'\delta' - 3\delta^3 - \delta^4
 \end{aligned}$$

and so on.

5. Thus suppose (p_4, q_4, r_4, s_4) are given, we have δ determined by a differential equation (not linear) of the third order; and δ being known, p_3, q_3, r_3 are also known; then γ is determined by a differential equation (not linear) of the second order, and γ being known, p_2, q_2 are also known; then β is determined by a differential equation (not linear) of the first order, and β being known, p_1 , that is, α is also known.

Comparing the last equations of each set, that is, the equations for the determination of $\gamma, \beta, \delta, \dots$ respectively, it will be observed that they depend on the single series of derivatives

$$-1, \phi, \phi - \phi', \phi'' - 3\delta\phi' + \phi', \phi''' - 4\delta\phi'' + 6\phi'\phi' - 3\delta\phi' - \phi^4,$$

where, calling the successive terms $A, B, C, D, E, \dots$, we have

$$B = A' - \phi A, \quad C = B' - \phi B, \quad D = C' - \phi C.$$

6. *Direct Order.* Considering the product

$$\left(\frac{d}{dx} + \alpha\right)\left(\frac{d}{dx} + \beta\right)\left(\frac{d}{dx} + \gamma\right)\left(\frac{d}{dx} + \delta\right)\cdots$$

[Paper 851 continues on subsequent pages.]